

ECEN 215, Spring 2024

Electrical Circuit Theory

Class 12: Capacitor, Inductor, Transient

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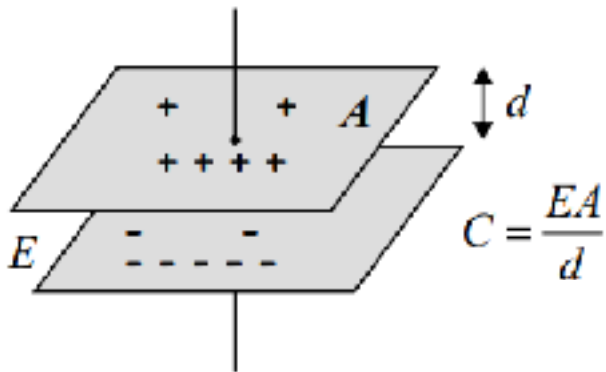


- Capacitors
- Inductors
- Transient circuit analysis
 - Natural/Step Response R-C/R-L Circuits
- Next Lecture

Capacitor



- Capacitor is a passive electrical component consists of two electrical conductors separated by an insulator and the capacitance is related to area of the conductors and the spacing between them, where ϵ is the dielectric primitivity of the insulator
- Capacitance (C) is measured in Farad (F) named after Michael Farady



Michael Farady

Capacitor



- Capacitors store energy in the form of an electric field between two charged plates (sort of like it is “storing voltage”)

Energy Stored in a Capacitor

Work done by a battery/power supply in “charging” a capacitor is stored as **electrical potential energy** in an **electric field** which exists between the charged capacitor plates.

This **electrical potential energy** is released when the capacitor is **discharged**, e.g., by connecting both plates of the capacitor to a light bulb.

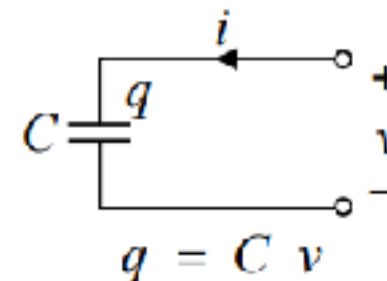
$$E = \frac{1}{2} Q V = \frac{1}{2} C V^2 = \frac{1}{2} \frac{Q^2}{C}$$

E = energy stored in capacitor/ joules (J)

Q = charge stored on capacitor plates/ coulombs (C)

V = potential difference between capacitor plates/ volts (V)

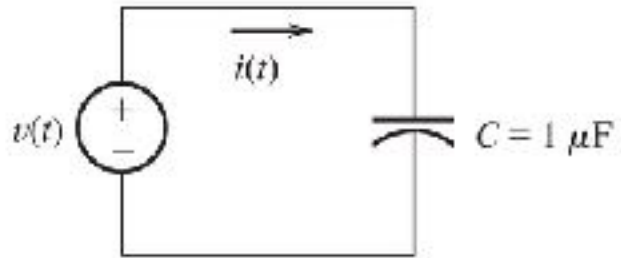
C = capacitance of capacitor/ farads (F)



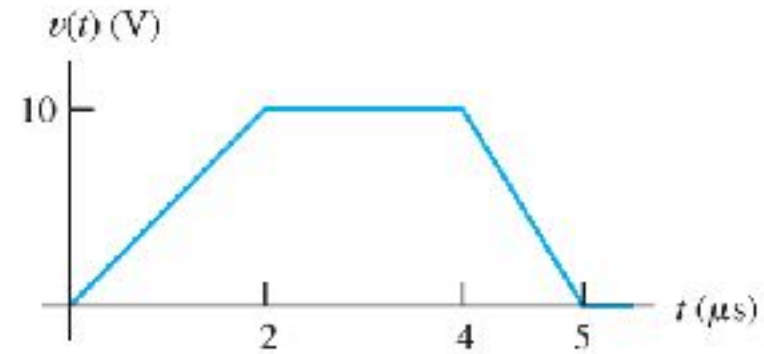
$$\begin{aligned} i &= \frac{dq}{dt} \\ &= \frac{d(Cv)}{dt} \\ &= C \frac{dv}{dt} \end{aligned}$$

$$\left[E = \frac{1}{2} C v^2 \right]$$

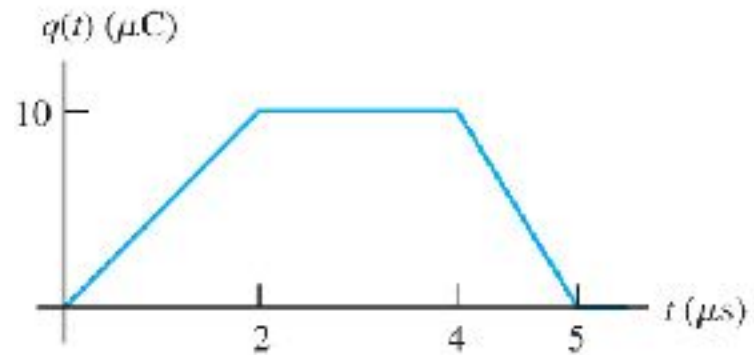
Example 1) Capacitor Current



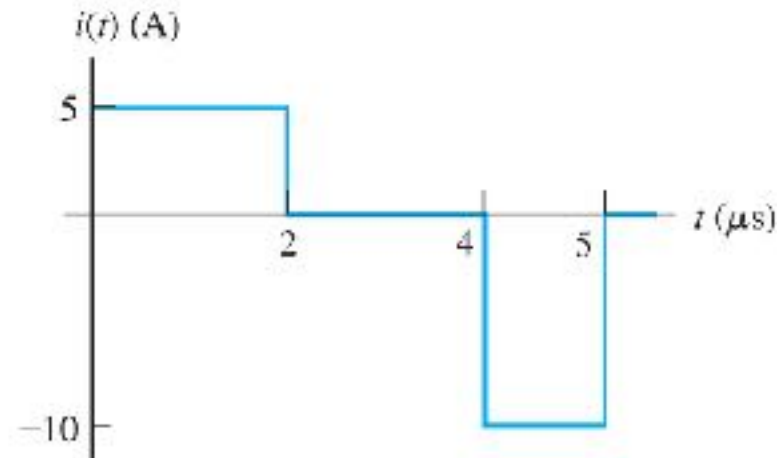
(a)



(b)



(c)

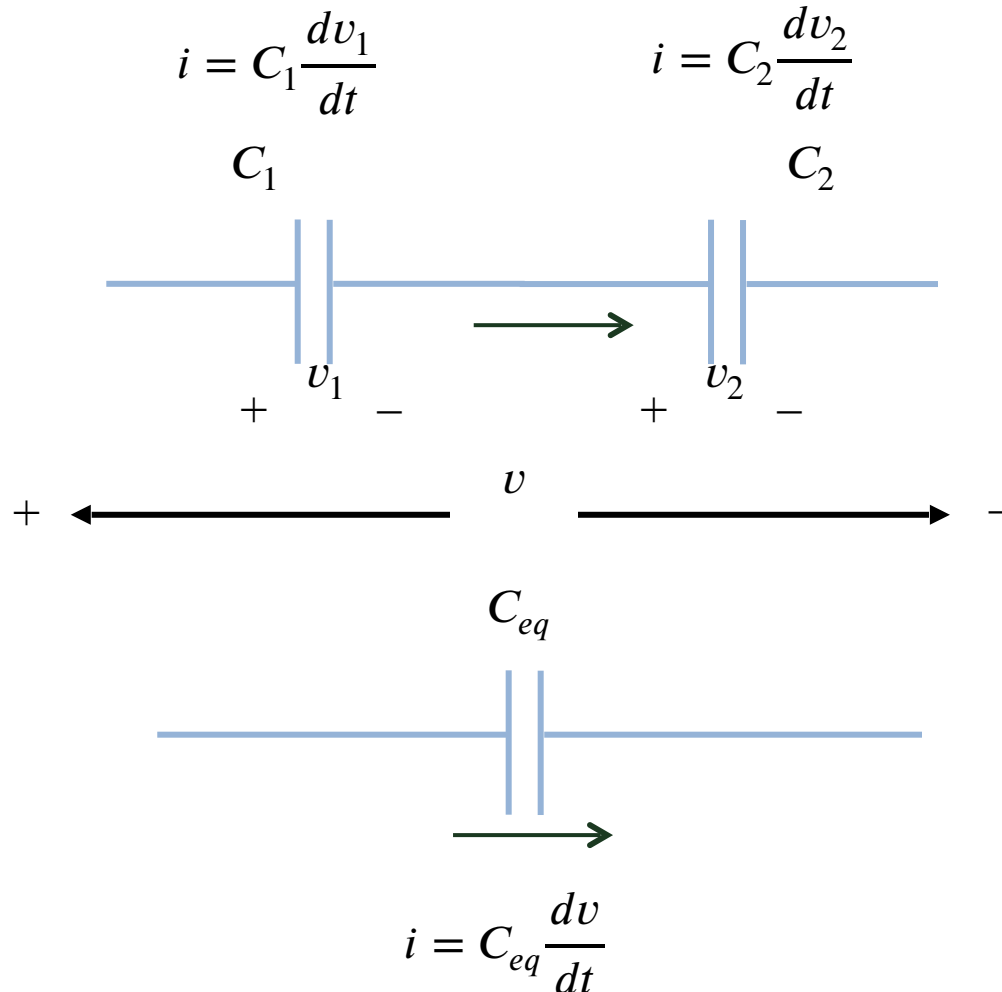


(d)

Capacitors in Series



- Capacitors in series combine like resistors in parallel.



Using KVL: $v = v_1 + v_2$

Differentiating: $\frac{dv}{dt} = \frac{dv_1}{dt} + \frac{dv_2}{dt}$

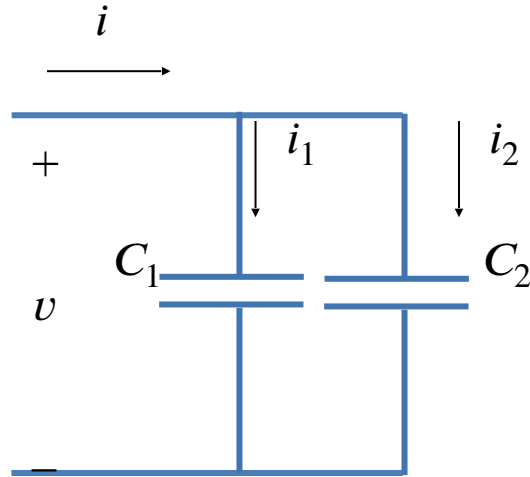
Substituting: $\frac{i}{C_{eq}} = \frac{i}{C_1} + \frac{i}{C_2}$

$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2}$$

Capacitors in Parallel



- Capacitors in parallel combine like resistors in series.



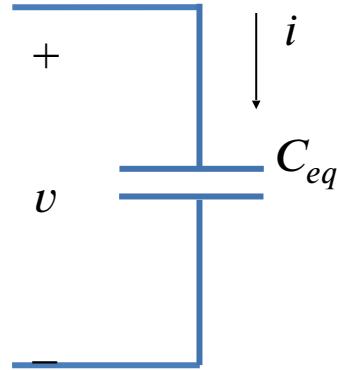
$$i_1 = C_1 \frac{dv}{dt}$$

$$i_2 = C_2 \frac{dv}{dt}$$

Using KCL: $i = i_1 + i_2$

Substituting:

$$\begin{aligned} C_{eq} \frac{dv}{dt} &= C_1 \frac{dv}{dt} + C_2 \frac{dv}{dt} \\ &= (C_1 + C_2) \frac{dv}{dt} \end{aligned}$$



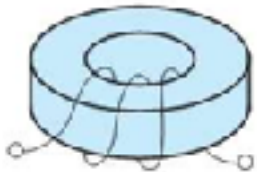
$$i = C_{eq} \frac{dv}{dt}$$

$$C_{eq} = C_1 + C_2$$

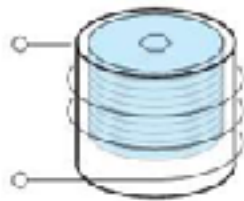
Inductor



- An inductor, also called a coil, is a passive two-terminal electrical component that stores energy in a magnetic field. It typically consists of an insulated wire wound into a coil.
- Inductance (L) is measured in Henry (H) named after Joseph Henry



(a) Toroidal inductor



(b) Coil with an iron-oxide slug that can be screwed in or out to adjust the inductance

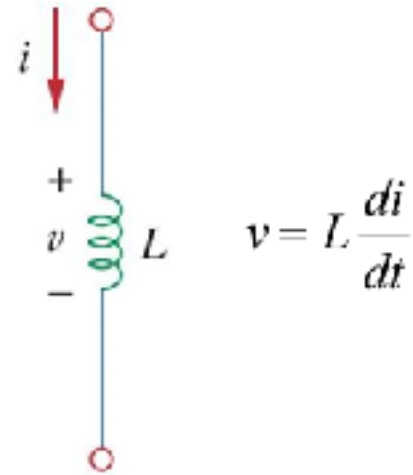
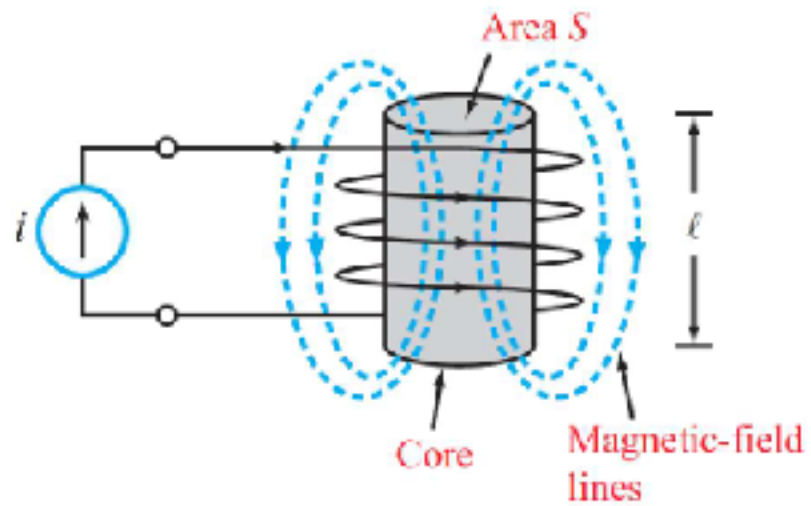


Joseph Henry

Inductor



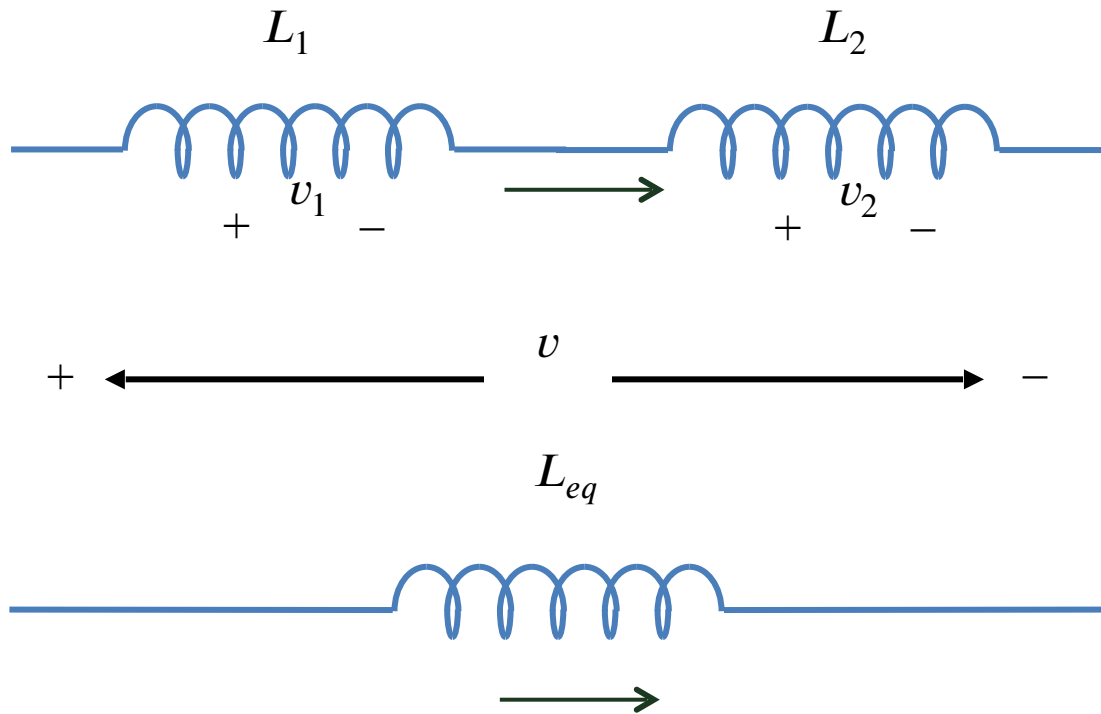
- Inductors store energy in the form of an magnetic field (sort of like it is “storing current”)



Inductors in Series



- Inductors combine in series exactly like resistors.



$$v_1 = L_1 \frac{di}{dt} \quad v_2 = L_2 \frac{di}{dt}$$

$$v = v_1 + v_2 = (L_1 + L_2) \frac{di}{dt}$$

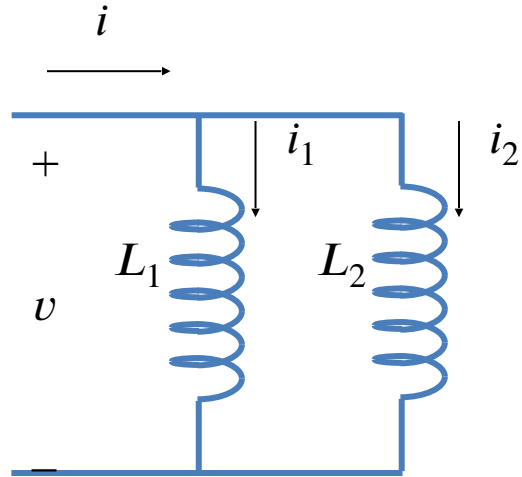
$$v = L_{eq} \frac{di}{dt}$$

$$L_{eq} = L_1 + L_2$$

Inductors in Parallel



- Inductors combine in parallel exactly like resistors.

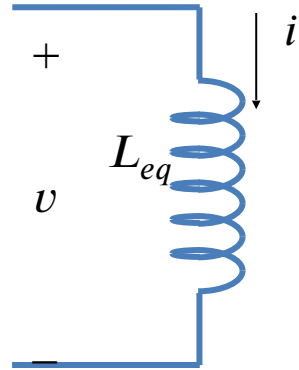


$$v = L_1 \frac{di_1}{dt}$$

$$v = L_2 \frac{di_2}{dt}$$

Using KCL: $i = i_1 + i_2$

Differentiating: $\frac{di}{dt} = \frac{di_1}{dt} + \frac{di_2}{dt}$



$$v = L_{eq} \frac{di}{dt}$$

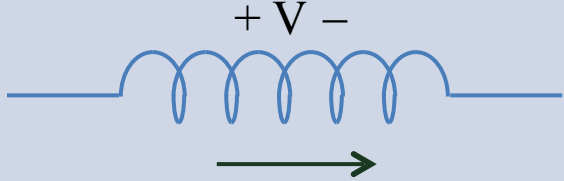
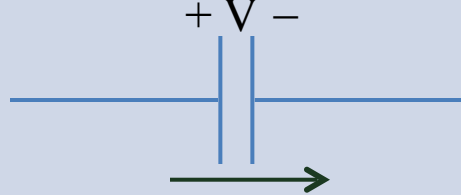
Substituting: $\frac{v}{L_{eq}} = \frac{v}{L_1} + \frac{v}{L_2}$

$$\boxed{\frac{1}{L_{eq}} = \frac{1}{L_1} + \frac{1}{L_2}}$$

Inductors vs. Capacitors



- Inductors and capacitors are passive devices which store (and release) energy (energy is not created/generated by a passive device).

	Inductor	Capacitor
Circuit Symbol		
Fundamental I/ V relationship		
Units	L=inductance (Henrys)	C=capacitance (Farads)



Properties of Inductors and Capacitors

- From the fundamental I/V relationship we can infer several properties of inductors and capacitors.

	Inductor	Capacitor
Fundamental I/V relationship		
Instantaneous changes	The current in an inductor cannot change instantaneously. To do so would cause an infinite voltage to appear.	The voltage across a capacitor cannot change instantaneously. To do so would cause an infinite current to flow.
Steady State (when things are not changing with time)	When a circuit with an inductor reaches steady state, the voltage across the inductor will be zero. → An inductor acts like a short circuit in steady state.	When a circuit with a capacitor reaches steady state, the current through the capacitor will be zero. → An capacitor acts like an open circuit in steady state.

Inverse to Fundamental I–V Relationships



- From the fundamental I–V relationships we can also derive inverse relationships.

Inductors

$$v = L \frac{di}{dt}$$

$$\int_{t_o}^t v(u) du = L \int_{t_o}^t \frac{di}{du} du$$

$$\int_{t_o}^t v(u) du = L(i(t) - i(t_o))$$

$$i(t) = i(t_o) + \frac{1}{L} \int_{t_o}^t v(u) du$$

Capacitors

$$i = C \frac{dv}{dt}$$

$$\int_{t_o}^t i(u) du = C \int_{t_o}^t \frac{dv}{du} du$$

$$\int_{t_o}^t i(u) du = C(v(t) - v(t_o))$$

$$v(t) = v(t_o) + \frac{1}{C} \int_{t_o}^t i(u) du$$

Power and Energy in Inductors and Capacitors



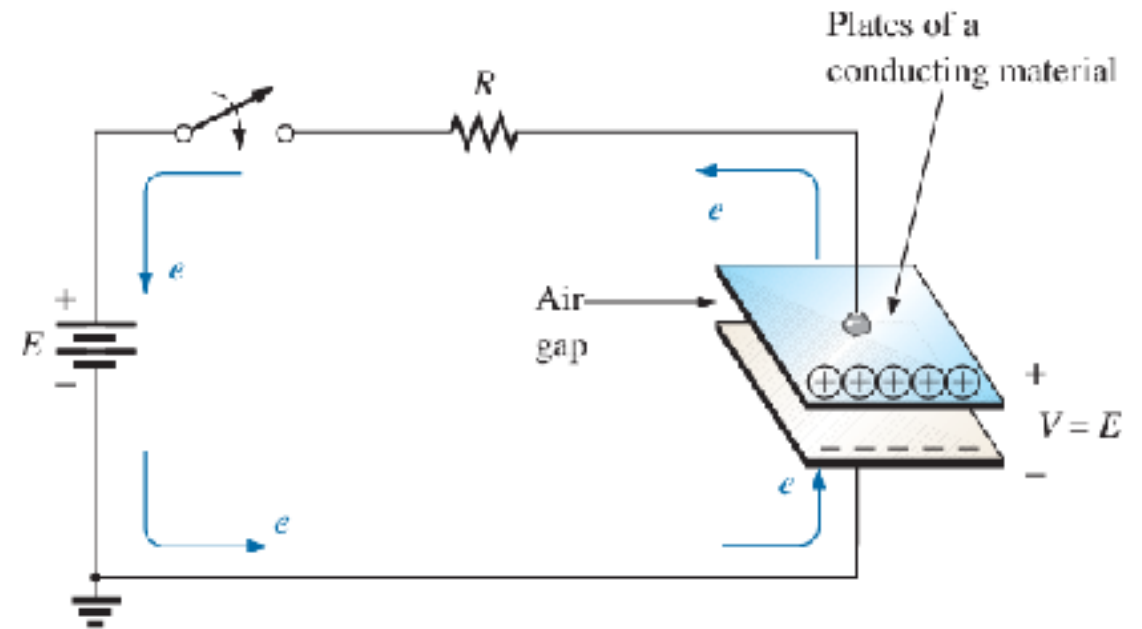
- From the fundamental I–V relationship we can infer several properties of inductors and capacitors.

	Inductor	Capacitor
Fundamental I–V relationship		
Power		
Energy		

Capacitor Circuits in Transient: How it works



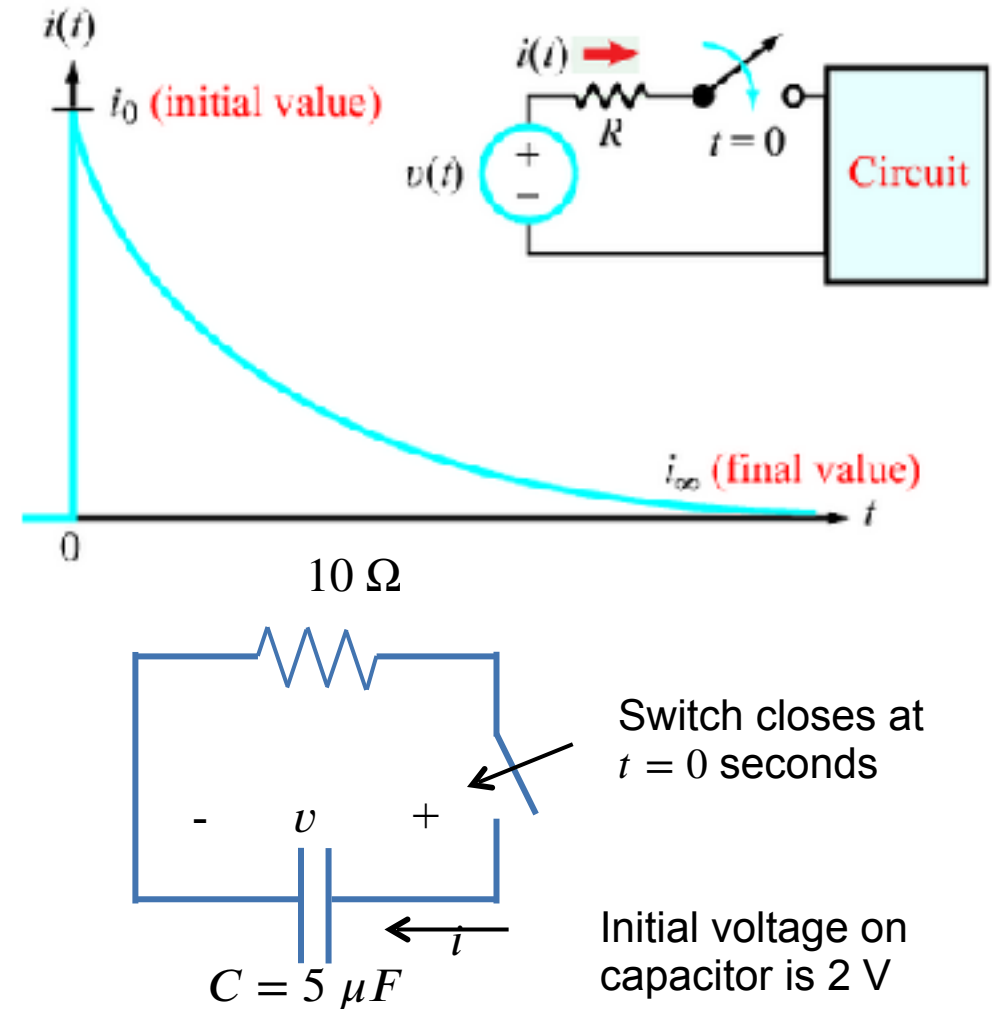
- Now, we will look at how circuits change over time (Transient)
- Capacitor circuit when switch is closed
 - electrons are drawn from the upper plate through
 - the resistor to the positive terminal of the battery
 - Surge in current limited in magnitude by R
 - a net positive charge on the top plate.
 - until the potential difference $V = E$



Transient Circuit Analysis



- We'll solve these
- (1) steady-state before the switch
($t < 0$) $\rightarrow$ C: O.C. and L is Sh.C
- (2) instantaneously after the switch
($t=0$)
- (3) steady-state after the switch
($t > 0$) $\rightarrow$ C: O.C. and L is Sh.C





Example 2 discharging RC Circuit

1. Steady-state before the switch:

$$v = 2 \text{ V}, i = 0 \text{ A}$$

$$\text{Energy stored in the capacitor} = \frac{1}{2}v^2C = 10 \mu\text{J}$$

2. Instantaneously after the switch:

$$v = 2 \text{ V}$$

$$i = -\frac{2\text{V}}{10\Omega} = -0.2 \text{ A}$$

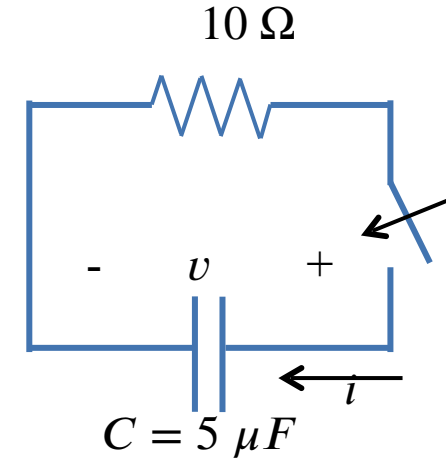
$$\text{Equation: } C \frac{dv(t)}{dt} = i(t) = -\frac{v(t)}{R}$$

3. In steady state after the switch:

$$i = 0 \text{ A}$$

$$v = 0 \text{ V}$$

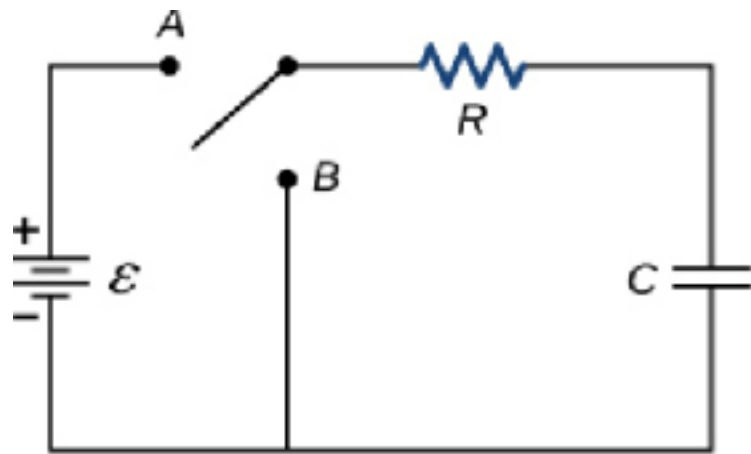
$$\text{Energy stored in the capacitor} = 0 \text{ J}$$



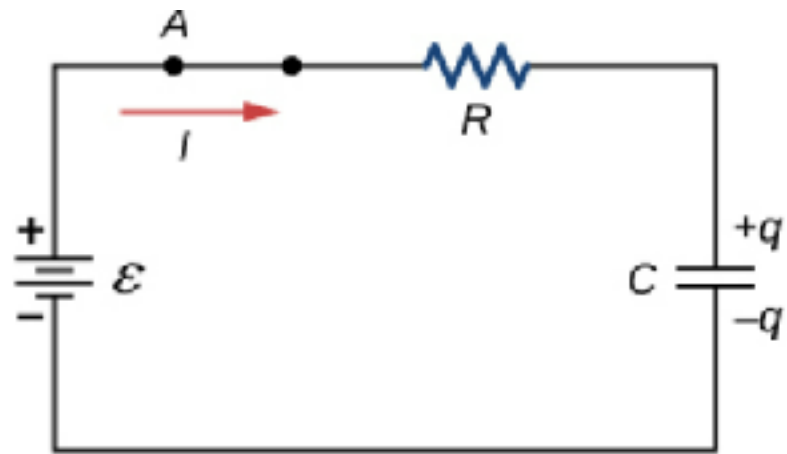
Switch closes at
 $t = 0$ seconds

Initial voltage on
capacitor is 2 V

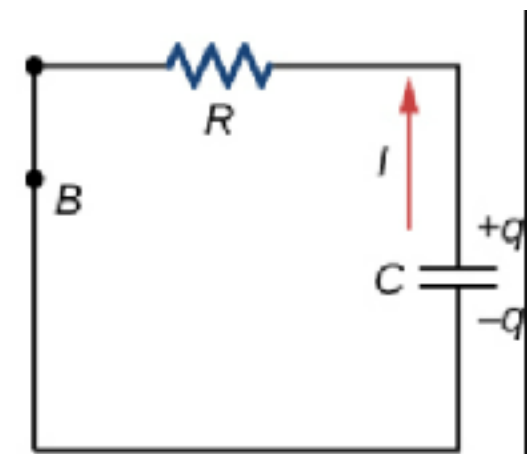
Capacitor Charging and discharging



(a) Original circuit

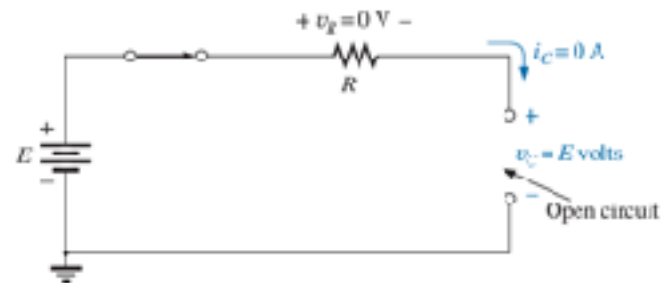


(b) Charging capacitor



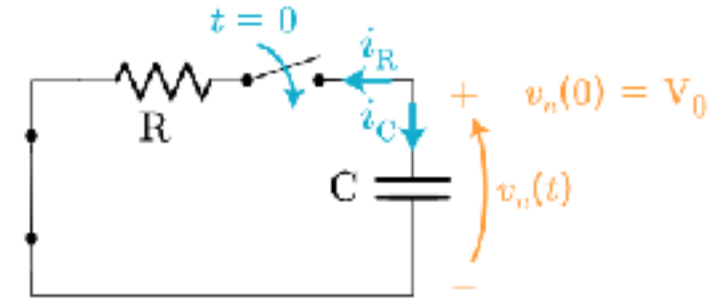
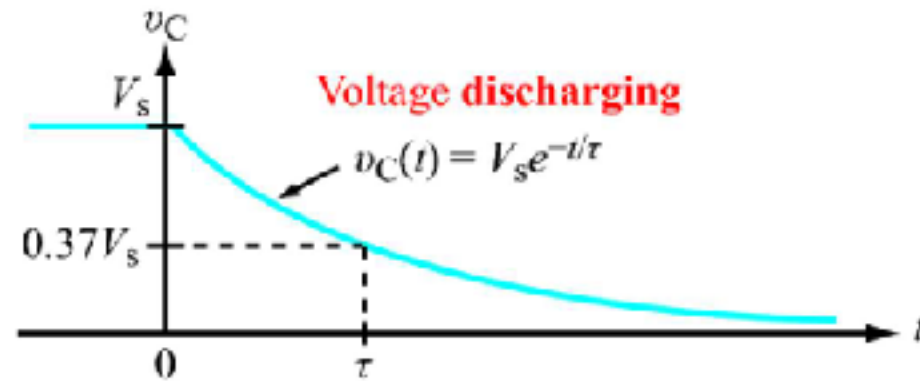
(b) Discharging capacitor

A capacitor can be replaced by an open-circuit equivalent once the charging phase in a dc network has passed.



Demonstrating that a capacitor has the characteristics of an open circuit after the charging phase has passed.

Natural Response (discharging) capacitor

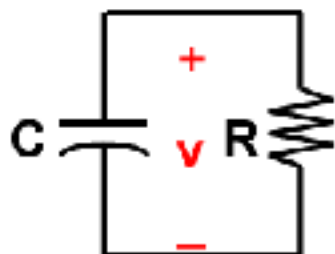


Time constant: $\tau = RC$

Natural Response RL and RC



RC Circuit



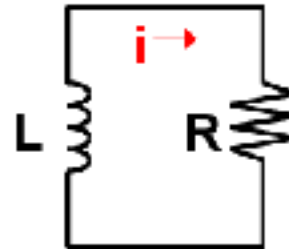
- Capacitor voltage cannot change instantaneously

$$v(0^-) = v(0^+)$$

$$v(t) = v(0)e^{-t/\tau}$$

- time constant $\tau = RC$

RL Circuit



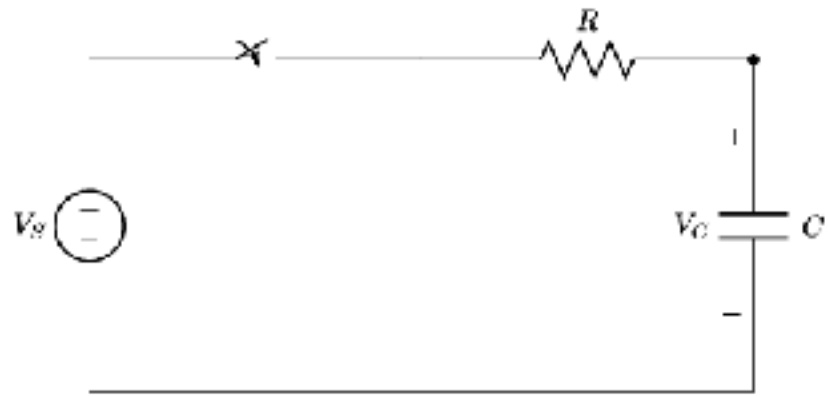
- Inductor current cannot change instantaneously

$$i(0^-) = i(0^+)$$

$$i(t) = i(0)e^{-t/\tau}$$

- time constant $\tau = \frac{L}{R}$

Capacitor Charging



$$\frac{V_S - V_C}{R} = i_C$$

$$\frac{V_S - V_C}{R} = C \frac{dV_C}{dt}$$

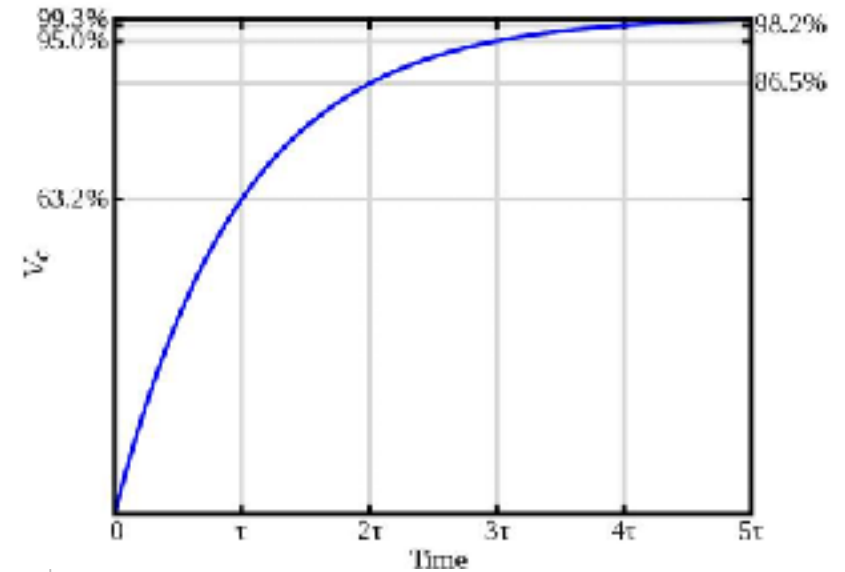
$$\frac{dt}{RC} = \frac{dV_C}{V_S - V_C}$$

$$\int_0^t \frac{1}{RC} dt = \int_0^t \frac{1}{V_S - V_C} dV_C$$

$$\frac{t}{RC} = -\ln(V_S - V_C) + C_1$$

$$C_2 e^{\frac{-t}{RC}} = V_S - V_C$$

$$V_C(t) = V_S(1 - e^{\frac{-t}{RC}})$$





What You Need to Know Today

- Capacitors
 - Fundamental equation: $i = C \frac{dv}{dt}$, where C is capacitance in Farads
 - Add in parallel, combine in series the way resistors do in parallel
 - In steady-state, they act like an open circuit
 - Energy stored is Cv^2
- Inductors
 - Fundamental equation: $v = L \frac{di}{dt}$, where L is inductance in Henries
 - Combine in series and parallel the same way as resistors
 - In steady-state, they act like a short circuit
 - Energy stored is Li^2
- Transient circuit solutions
 - We are solving for v and i as a function of time, $v(t)$ and $i(t)$
 - Write the differential equations using KVL and KCL
 - Next class we will start to work on solving them

Next Lecture



- Phasor